



Domain : Open Innovation

Title : VoltCustodian

Team Name : MECHA-ME



VOLTCUSTODIAN

**IOT-SUPERVISED POWER
PROTECTION & MONITORING
STATION**

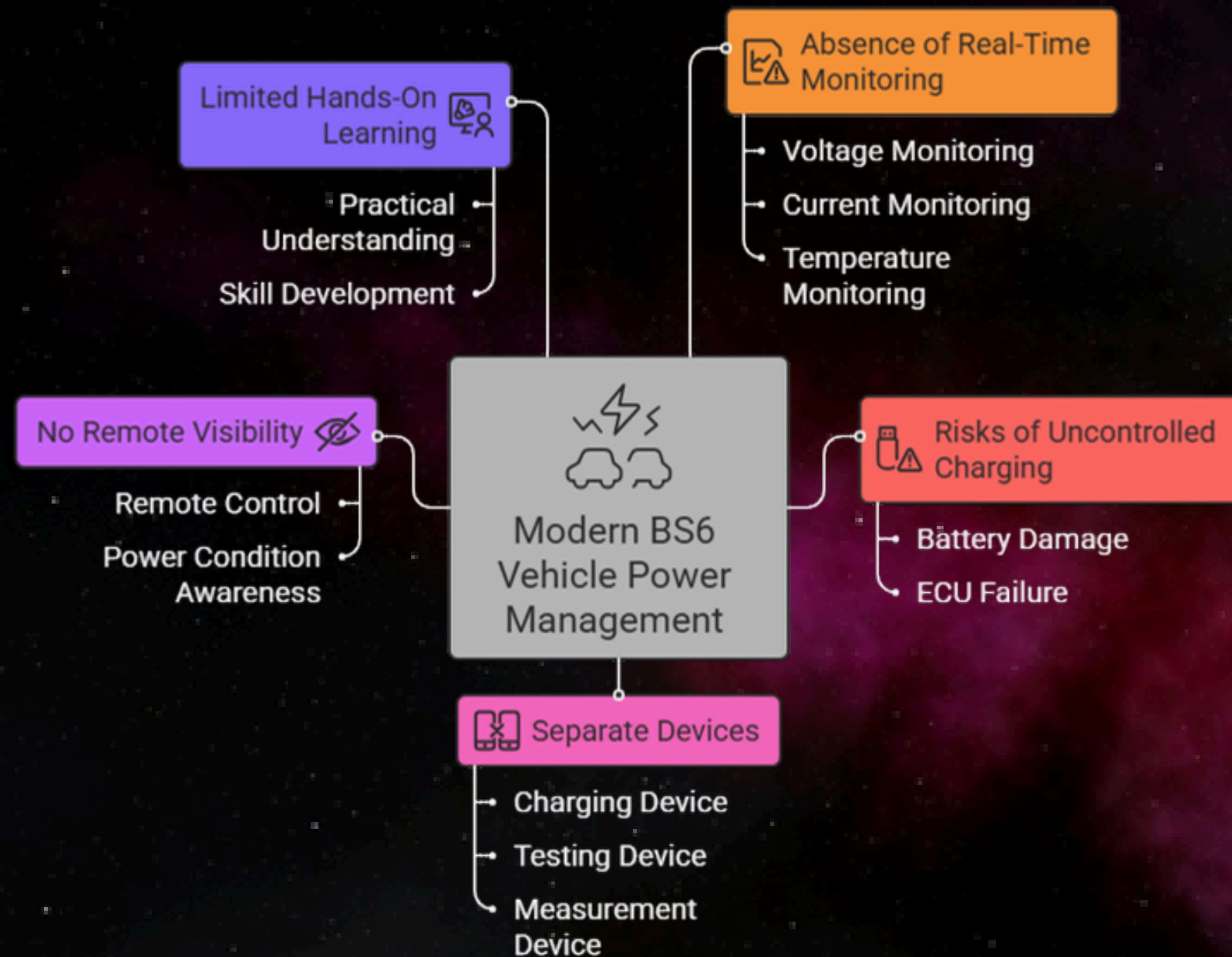


**MECHA-ME #
FUTURETECH**



PROBLEM STATEMENT

IDENTIFYING KEY CHALLENGES WE FACE



Modern BS6 vehicles rely heavily on electronic control systems that demand safe, stable, monitored power and ersatile battery charging solution. Existing battery charging and circuit testing methods operate without real-time supervision, increasing the risk of component damage, unsafe operation, and inaccurate diagnostics. Educational environments also lack integrated systems that demonstrate live power behavior, limiting practical understanding and skill development.

Key Problem :

- Limited Battery Compatibility – Most chargers work for only one type/voltage.
- Safety Risks – Overcharge, overcurrent, and overheating hazards.
- No Real-Time Monitoring – Users cannot track voltage, current, or temperature.
- No IoT/Remote Control – Cannot monitor or manage multiple devices remotely.
- Limited Educational & Industrial Use – No real-time data for learning or testing.
- Low Scalability – Cannot adapt to evolving EV battery standards.

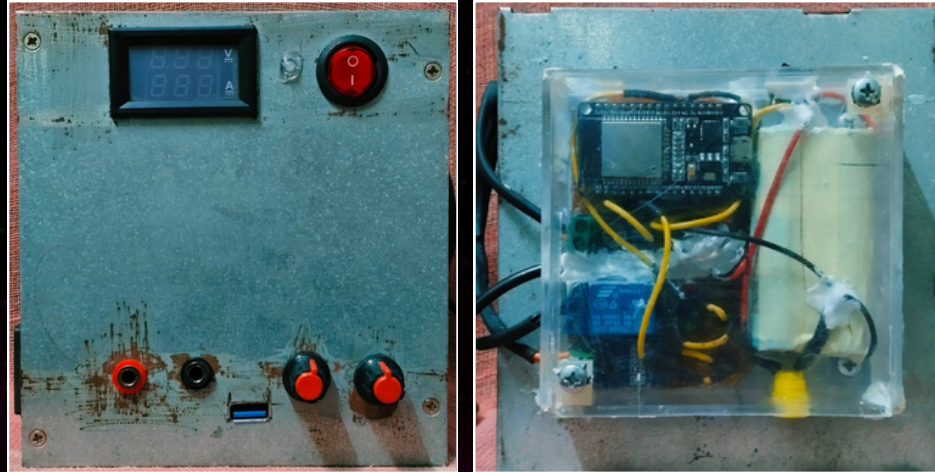


PROPOSED SOLUTION

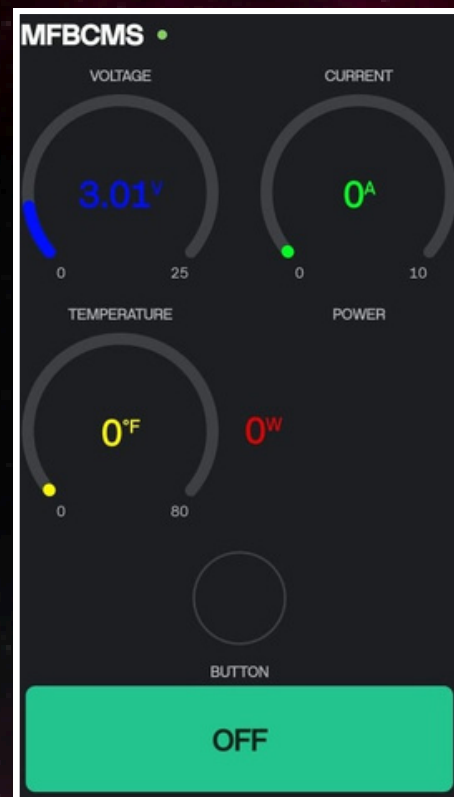
A PATH TO INNOVATION



HARDWARE INTERFACE



SOFTWARE INTERFACE



- IoT-enabled multifunctional system for charging, testing & bench supply.
- ESP32 with sensors (ACS712, DS18B20, Voltage divider) for real-time data.
- Relay control for safe ON/OFF charging operation.
- Live monitoring of voltage, current, power, and temperature.
- Blynk IoT app for remote control and data visualization.
- Safety features: overcurrent, overvoltage, overtemperature protection.
- Cost-effective & compact solution for workshops and education.



SYSTEM ARCHITECTURE



THE BUSINESS MODEL CANVAS

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
- Battery Manufacturers & Suppliers - Power Electronics & SMPS Suppliers - Renewable energy providers (solar, grid) - Automotive Workshops & Service Centers - Utility companies / power distributors - IoT & cloud service providers	- BS6 battery smart charging system development - Power monitoring & load management - Software & IoT platform maintenance - Data analytics & optimization - System testing, deployment & upgrades - Customer support & system monitoring KEY RESOURCES - Charging hardware & controllers - IoT sensors and communication modules - Cloud infrastructure & databases - Software platform & mobile app - Skilled engineering & R&D team - Data analytics & AI models	- Smart, safe, and efficient BS6 battery charging - Reduced charging time & energy wastage - Real-time monitoring and control - Optimized power usage & grid protection - Improved battery life - Scalable solution for home, fleet & public charging	- Self-service via mobile/web app - Automated alerts & notifications - Technical support & maintenance services - Long-term subscriptions for fleets - Continuous software updates CHANNELS - Mobile application - Web dashboard - Direct sales to fleet operators - Direct Sales to Automotive Workshops - Automotive Spare Parts & Equipment Dealers	- Automotive Service Centers & Workshops - Industrial Maintenance Departments - Charging station operators - Residential societies - Corporate campuses & smart cities - Public infrastructure providers
COST STRUCTURE - Hardware manufacturing & procurement - Software development & cloud services - Maintenance & technical support - R&D and system upgrades - Marketing & partnerships - Operational and deployment costs		REVENUE STREAMS - Smart charging subscriptions - Licensing software to charging operators - Fleet management services - Installation & maintenance fees - Premium analytics & energy optimization services		

System Architecture

1. Power Conditioning Unit :-

- AC-DC conversion
- Voltage & current regulation

2. Smart Control Unit :-

- Microcontroller / Embedded system
- Executes charging logic and safety rules

3. Sensor Layer :-

- Voltage, current, temperature sensors
- Continuous real-time monitoring

4. Communication Layer :-

- IoT module (Wi-Fi / GSM / Bluetooth)
- Data transmission to cloud/dashboard

5. User Interface :-

- Mobile app / Web dashboard
- Displays charging status, alerts, and usage data

6. EV Battery Interface :-

- Controlled and safe energy delivery to BS6 vehicle batteries





TECHNICAL STACK



Hardware Stack :-

- EV Battery Pack (12V–48V)
- AC–DC Power Supply (SMPS)
- Microcontroller (ESP32 / Arduino / STM32)
- Voltage, Current & Temperature Sensors
- Relay / MOSFET Switching Circuit
- Protection Circuits (Overvoltage, Overcurrent)
- LCD / OLED Display

Software Stack :-

- Embedded C / Arduino IDE
- Control Algorithms for Smart Charging
- Power Management Logic
- Fault Detection & Safety Handling

IoT & Communication Stack :-

- Wi-Fi / MQTT / HTTP Protocols
- Cloud Platform (Blynk / Firebase / ThingSpeak)
- Real-time Data Monitoring Dashboard
- Alert & Notification System

Data & Analytics :-

- Charging Data Logging
- Usage & Power Consumption Analysis
- Performance Optimization Insights

APP WORKING VIDEO LINK :

https://youtu.be/i9uClPnuwRM?si=_d6yZ01gKG1ma3kl



IMPLEMENTATION PLAN



External Entities

- User (Owner)
- Power Grid / Energy Source

Process Flow

1. User Input

- User starts/stops charging and views status via mobile app / web dashboard

2. Data Acquisition

- Sensors collect:
 - Voltage
 - Current
 - Temperature
 - Battery State of Charge (SoC)

3. Smart Control Processing

- Microcontroller processes sensor data
- Applies charging algorithms and safety rules
- Decides optimal charging rate

4. Power Management

- Regulates power flow from grid to EV battery
- Prevents overcharging, overheating, and overload

5. Cloud & IoT Communication

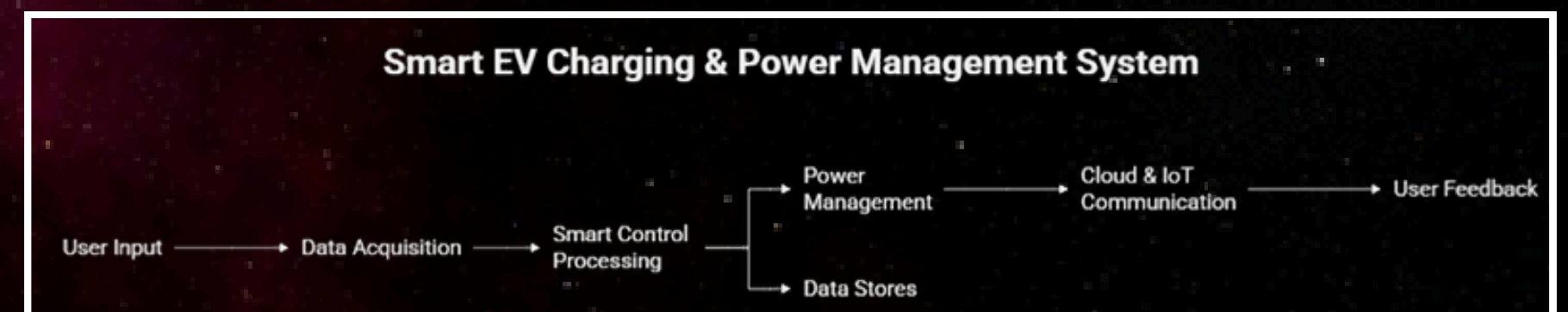
- Charging data sent to cloud server
- Stores logs, usage history, and alerts

6. User Feedback

- Real-time status, notifications, and reports shown to user

Data Stores

- Cloud Database
- (Charging history, energy consumption, system logs)





THE CONCLUSION

WHY VOLT CUSTODIAN MATTERS



Why VoltCustodian Matters :

- This project demonstrates a practical, scalable, and future-ready approach to smart BS6 Battery charging and power management.
- Our solution combines safety, efficiency, and intelligent control to address real-world charging challenges.
- It is designed to adapt to growing adoption, evolving energy needs, and smart grid ecosystems.
- With future enhancements like AI optimization, renewable integration, and V2G, the impact can extend beyond charging to smart energy management.
- We believe this solution has strong potential for real-world deployment and commercial scalability.

Vision :

- To enable safe, intelligent, and energy-efficient BS6 Smart Battery charging through smart power management, real-time monitoring, and scalable technology—supporting sustainable mobility and future-ready infrastructure.

One-Line Power Statement:

“VoltCustodian is not just a charger—it’s a smart energy solution for the future of electric mobility.”

THANK
YOU

